Afifa Tasneem Quanita

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LinkedIn | GitHub | ResearchGate

Objective

To pursue graduate study in Computer Science and Engineering with a focus on advancing research in artificial intelligence, machine learning, natural language processing, and cybersecurity. The objective is to contribute to the development of secure, reliable, and intelligent solutions for emerging cybersecurity challenges and to conduct impactful research that addresses real-world security and technological problems.

Work Experience

Lecturer – Varendra University *August 2025 – July 2026*
Rajshahi, Bangladesh

– **Course Title: Database Management System** *Summer 2025, Spring 2026*
– **Course Title: Computer Fundamentals** *Summer 2025, Spring 2026*
– **Course Title: E-commerce and Web Programming** *Spring 2026*

Education

Rajshahi University of Engineering & Technology *Jan 2020 – Jun 2025*
B.Sc. in Computer Science & Engineering • Rajshahi, Bangladesh

– CGPA: **3.55**/4.00 (Top 18%).

Rajshahi College *2019*
H.S.C. in Science • Rajshahi, Bangladesh

– GPA: **5.00**/5.00.
– Got General Stipend.

Rajshahi Government Girls' High School *2017*
S.S.C. in Science • Rajshahi, Bangladesh

– GPA: **5.00**/5.00.
– Got General Stipend.

Test Scores

IELTS Academic: Overall **8.0** (Listening: 9.0, Reading: 8.0, Writing: 7.0, Speaking: 7.0)

Skills

Programming Languages: C, C++, Python, JAVA
Web Technologies: JavaScript, HTML, CSS
Database Systems: MySQL, MongoDB
Data Science & Machine Learning: Pytorch, Keras, Tensorflow

Research Experience

Young Learners Research Lab, Rajshahi University of Engineering & Technology *June 2023 – June 2025*
Executive Member

– Assisted in research, delivering a research presentation, demonstrating exceptional dedication and contributing to ongoing efforts in the Computer Science & Engineering Department.

Undergraduate Thesis *July 2024 – June 2025*
Enhancing Cyber Threat Intelligence Using Transformer Models for Text Classification

– Developed a robust methodology for automated text classification to enhance Cyber Threat Intelligence (CTI), addressing evolving threats like malware, phishing, and ransomware.

Review Paper: Cybersecurity in Internet of Things (IoT)
Technical Writing course • Group review project • First research paper • Unpublished *Preprint*

– Co-authored *Cybersecurity in Internet of Things (IoT), A Comprehensive Review*, surveying IoT architecture and the security challenges affecting the perception, network, processing, and application layers.
– Synthesized prior research on layer-specific attacks, cybersecurity requirements, and countermeasures including authentication, encryption, access control, IDS/IPS, anomaly detection, secure communication protocols, and physical-security measures.

Publications

C = Conference, J = Journal, P = Patent, S = In Submission, T = Thesis; own name is in bold.

Peer-Reviewed Conference Papers

[C1] **A. T. Quanita** and M. A. Y. Srizon, **Enhancing Cyber Threat Intelligence Using Transformer Models for Text Classification**. *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, Chittagong, Bangladesh, 2026, pp. 1–5. DOI: 10.1109/QPAIN69676.2026.11546420.

[C2] S. Agarwala, **A. T. Quanita**, Onik Jahan Sagor, **Leveraging Deep Learning and Feature Extraction for Robust Anomaly Detection in Network Traffic**. *27th International Conference on Computer and Information Technology*, Cox's Bazar, Chattogram, Bangladesh, 2024. DOI: 10.1109/ICCIT64611.2024.11022455.

[C3] S. Agarwala, M. S. Ali, **A. T. Quanita**, **Precision Death Forecasting of Dengue Outbreaks in Bangladesh by Blending Statistical and Machine Learning Models**. *2024 IEEE International Conference on Power, Electrical, Electronics and Industrial Applications*, Rajshahi, Bangladesh, 2024, pp. 981–986. DOI: 10.1109/PEEIACON63629.2024.10800576.

[C4] S. Agarwala, A. S. M. S. Sajid, **A. T. Quanita**, M. S. Ali and S. M. M. Hasan, **Enhancing Dengue Outbreak Prediction in Bangladesh: A Comparative Study of Advanced Predictive Models**. *2024 IEEE International Conference on Computing, Applications and Systems (COMPAS)*, Cox's Bazar, Bangladesh, 2024, pp. 1–6. DOI: 10.1109/COMPAS60761.2024.10796965.

[C5] S. Agarwala, M. S. Ali, **A. T. Quanita**, **A Cross-Analyzing Approach to Sentiment and Bias Detection in Social Media: Insights from Geopolitical Conflicts**. *27th International Conference on Computer and Information Technology*, Cox's Bazar, Chattogram, Bangladesh, 2024. DOI: 10.1109/ICCIT64611.2024.11021854.

Projects

Software Development Project: RUET Bus Manager
Android application • Second academic year of undergraduate studies *GitHub*

– Built an Android app called "RUET Bus Manager" using JavaScript in Android Studio. The app notifies users when their bus will arrive at their desired station, allowing them to set their route and time accordingly.

Web Development Project: University Website
Third academic year of undergraduate studies *GitHub*

– Built a university website using HTML, CSS, and JavaScript, focusing mainly on front-end development.

Professional Memberships

SAKURA Science Club *Jan 2018 – Present*
SSP ID–17105750

Excellence Bangladesh *Oct 2021 – Sep 2024*
Campus Team Secretary

RUET Debating Club *Sep 2024 – Jun 2025*
Debate Mentor

Certificates

- **2025:** Certificate of Appreciation for demonstration of exceptional dedication and achievemnt in research activities under Young Learners' Research Lab, Rajshahi University of Engineering & Technology (RUET).
- **2025:** Completed a Training on Python(Django) under Enhancing Digital Goverrntment and Economy (EDGE) Project of Bangladesh Computer Council, ICT Division.
- **2024:** 1st Runners up in ProCon 2024 Debating Contest Organized by IEEE Bangladesh Section in Collaboration with IEEE SB KUET Section.
- **2023:** 1st Runners Up at HULT Prize RUET 2024 – Team ANANSI for the idea "Door to Door".
- **2023:** Selected for Cohort A in Smart Unibator – University Innovation Hub Program, Rajshahi University of Engineering & Technology (RUET).
- **2023:** Certificate of Leadership from CSE Department, Rajshahi University of Engineering & Technology (RUET) for the second academic year of 2019–2020 session.
- **2022:** Certificate of Recognition for volunteering in the 4th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2022), organized by the faculty of Electrical and Computer Engineering, Rajshahi University of Engineering & Technology (RUET).
- **2018:** Certificate of Achievement – Japan-Asia Youth Exchange program in Science (SAKURA Exchange Program in Science), Okayama University, Japan.

Additional Information

Languages: Bangla (Native), English

Interests: Books, Debate, Poetry

References

Md. Azmain Yakin Srizon — *Thesis Supervisor*
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